



CAPSTONE PROJECT 2026

**CEPHFS+SMB HIGH
AVAILABILITY
+VIRTUAL OFFICE NETWORK**

NSCC
IT SYSTEMS MANAGEMENT & SECURITY

Project TEAM



MAHAN DIANATI



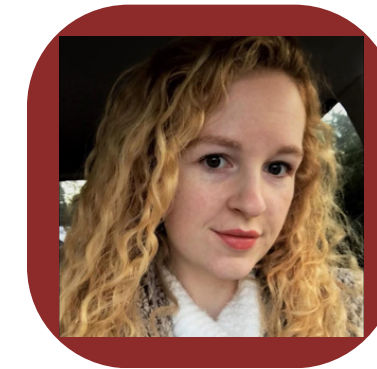
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Why **CEPH?**

Challenges in Modern Storage:

- Rapid growth of organizational data
- Single points of failure in centralized storage
- Limited scalability of traditional storage systems
- High operational cost of proprietary storage solutions

Why We Use Ceph:

- Fully distributed architecture
- Horizontally scalable
- Built-in redundancy and fault tolerance
- Open-source enterprise storage solution

Goals & **OBJECTIVES**

- Build Enterprise Storage Infrastructure
- Deploy Distributed Filesystem (Cephfs)
- Implement High-Availability SMB gateway (Samba & CTDB)
- Integrate with AD (EVE-NG)
- Simulate Corporate Network Environment
- Apply Security Best Practices

HARDWARE OVERVIEW

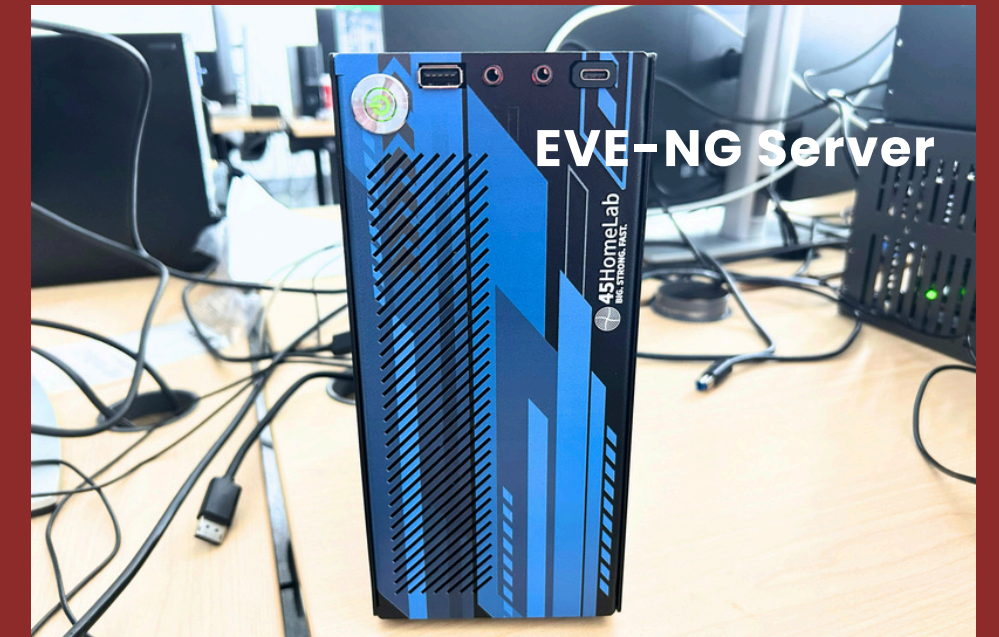


Rocky Linux servers:

- 4× Storage nodes
 - 2 SSDs
 - 5 HDDs
- 2× Gateway nodes

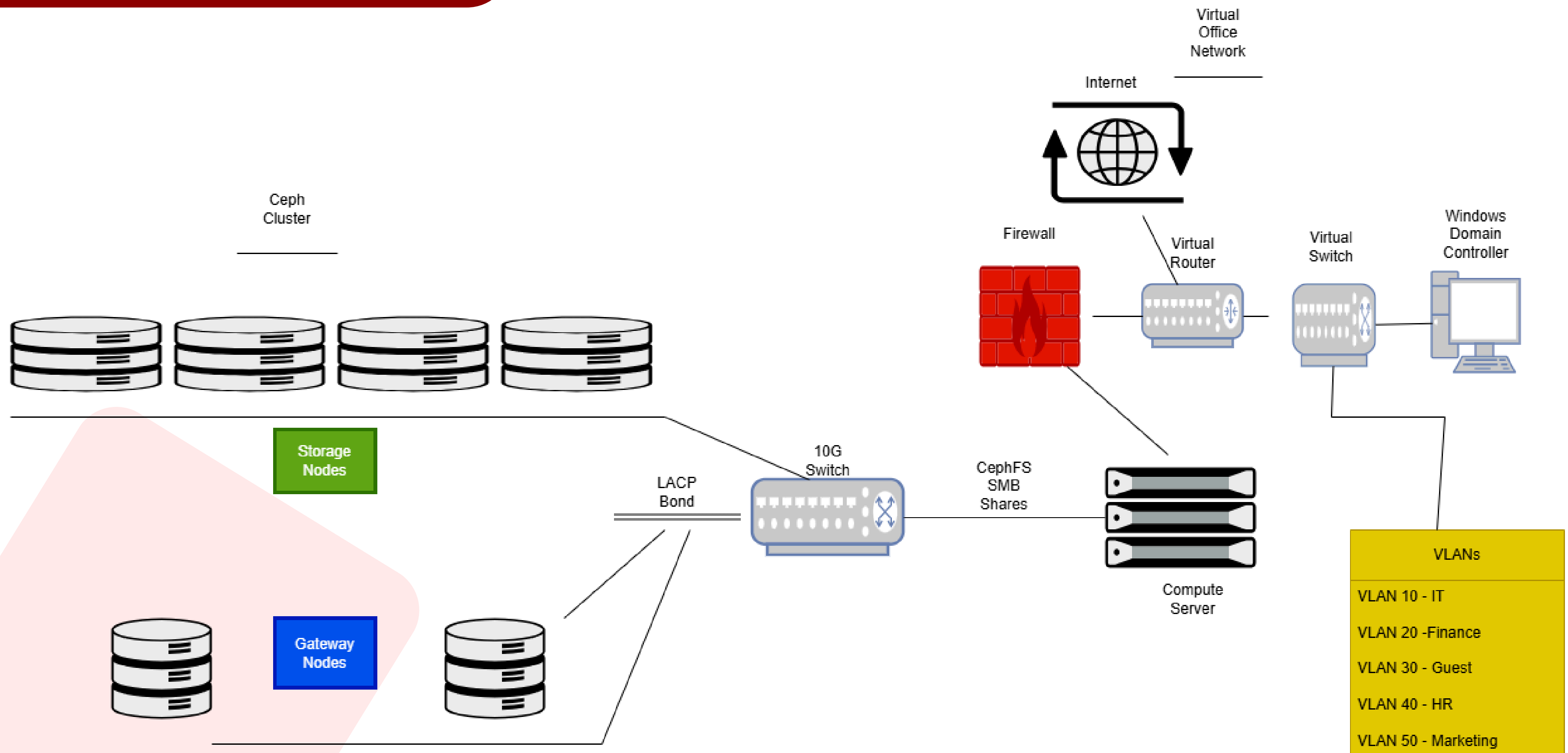


- 10GbE switch (TP-Link Omada)



- Compute Server (EVE-NG)

High-Level Architecture



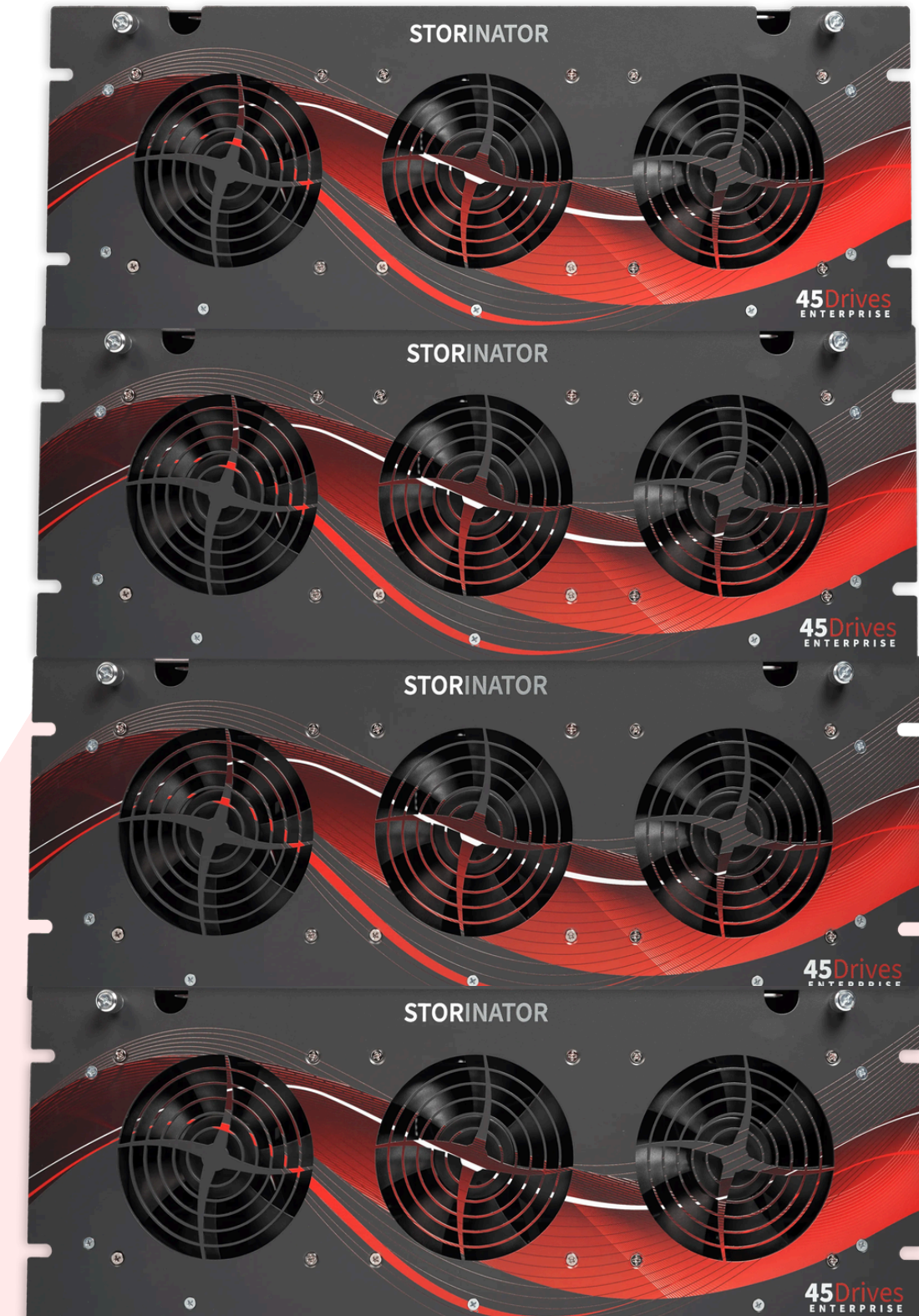
Ceph Architecture

	Hostname 	Service Instances 	Labels 	Status 	Model 	CPU's 	Cores 	Total Memory 	Raw Capacity 	HDD's 	Flash	NIC's
>	capstone-gw1 (192.168.45.1)	ceph-exporter: 1 crash: 1 mds: 1 node-exporter: 1	mds rgw _admin	available	Server (Storinator-MI4-Enhanced-S)	1	12	62.2 GiB	465.8 GiB	0	2	8
>	capstone-gw2 (192.168.45.2)	ceph-exporter: 1 crash: 1 mds: 1 node-exporter: 1	mds rgw _admin	available	Server (MS03-6L0-000)	1	12	62.2 GiB	465.8 GiB	0	2	6
>	capstone-sn1 (192.168.45.11)	ceph-exporter: 1 crash: 1 node-exporter: 1 osd: 7	_admin osd	available	Server (Storinator-AV15-Enhanced-AMD)	1	64	62.3 GiB	56.3 TiB	5	4	2
>	capstone-sn2 (192.168.45.12)	ceph-exporter: 1 crash: 1 mon: 1 osd: 7 node-exporter: 1	_admin mon osd	available	Server (Storinator-AV15-Enhanced-AMD)	1	24	125 GiB	56.3 TiB	5	4	4
>	capstone-sn3 (192.168.45.13)	mon: 1 mgr: 1 ceph-exporter: 1 crash: 1 alertmanager: 1 grafana: 1 prometheus: 1 osd: 7 node-exporter: 1	_admin mgr osd grafana mon	available	To be filled by O.E.M. (AV15)	1	26	124.9 GiB	56.3 TiB	5	4	4
>	capstone-sn4 (192.168.45.14)	ceph-exporter: 1 crash: 1 mgr: 1 mon: 1 node-exporter: 1 osd: 7	_admin mgr mon osd	available	To be filled by O.E.M. (AV15)	1	26	124.9 GiB	61.8 TiB	5	4	4

Ceph Cluster

DEPLOYMENT

- The cluster was deployed using cephadm
- Other storage nodes were added to the cluster
- Started appropriate daemons on specific nodes
- All logical & physical volumes and volume groups on disks were removed & filesystems were wiped
- Deployed OSDs on all storage nodes (5HDDs+2SSDs)



Ceph Cluster

VALIDATION

Status



```
[admin@capstone-sn3 ~]$ ceph -s
cluster:
  id:          be498ed4-1ca8-11f1-a5d1-0090faa5b49c
  health: HEALTH_OK

services:
  mon: 3 daemons, quorum capstone-sn3,capstone-sn4,capstone-sn2 (age 6d)
  mgr: capstone-sn3.kcftbq(active, since 8d), standbys: capstone-sn4.szqakv
  mds: 1/1 daemons up, 1 standby
  osd: 28 osds: 28 up (since 6d), 28 in (since 6d)

data:
  volumes: 1/1 healthy
  pools:   4 pools, 289 pgs
  objects: 3.46k objects, 13 GiB
  usage:   44 GiB used, 227 TiB / 227 TiB avail
  pgs:     289 active+clean
```

Tuning Steps

1. `mlnx_tune -p LOW_LATENCY_VMA`
 - a. Disabled `irqbalance` daemon, `nf_nat`, `nf_conntrack`, `nf_defrag_ipv4`, `nf_defrag_ipv6`
2. Increased buffer size (RX: max, TX: half capacity)
3. `tuned-adm profile network-latency`
4. Changed mtu size to 9000 on all interfaces and enabled jumbo frames on the switch

```
[admin@capstone-sn3 ~]$ ping -M do -s 8972 capstone-sn1
PING capstone-sn1 (192.168.45.11) 8972(9000) bytes of data.
8980 bytes from capstone-sn1 (192.168.45.11): icmp_seq=1 ttl=64 time=0.113 ms
8980 bytes from capstone-sn1 (192.168.45.11): icmp_seq=2 ttl=64 time=0.140 ms
8980 bytes from capstone-sn1 (192.168.45.11): icmp_seq=3 ttl=64 time=0.142 ms
^C
--- capstone-sn1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2051ms
```

CephFS

DEPLOYMENT

- Created CephFS
- GW1 as active mds and GW2 as standby
- Auto created metadata and data pools
- Turned off PG auto scale on the pools and set PG to 128 for data and to 32 for meta
- Set crush rules to ssd_rule for meta and to hdd_rule for data

	Name	Enabled	Created
▼	cephfs	✓	13 days ago

Details

Directories

Subvolumes

Subvolume groups

Snapshots

Snapshot schedules

Clients 3

Performance Details

Ranks

Rank	State	Daemon	Activity	Dentries	Inodes	Dirs	Caps
0	active	cephfs.capsto gw1.ssteph	Reqs: 0.4 /s	16	18	13	17
1 total							

Pools

Pool	Type	Size	Usage
cephfs.cephfs. data		70.8 TiB	0%
cephfs.cephfs. metadata		1.1 TiB	0%
2 total			

CephFS

SNAPSHOT STRATEGY

- Tiered Snapshot Strategy
- Short-term : Every Hour, Keeps 24 Snapshots
- Medium-Term : Daily Snapshots, Keeps 14 days
- Long-Term : Weekly Snapshot, Keeps 8 weeks
- Naming Scheme : hourly-2026-04-16-10, daily-2026-04-16, weekly-2026-W16

```
#!/bin/bash
set -euo pipefail

TARGET="/cephfs"
SNAPDIR="$TARGET/.snap"

# Safety checks
if [ ! -d "$TARGET" ]; then
    echo "Target path $TARGET does not exist"
    exit 1
fi

if [ ! -d "$SNAPDIR" ]; then
    echo "Snapshot directory $SNAPDIR does not exist or is not accessible"
    exit 1
fi

create_snap() {
    local snapname="$1"
    if [ ! -d "$SNAPDIR/$snapname" ]; then
        mkdir "$SNAPDIR/$snapname"
        echo "Created snapshot: $SNAPDIR/$snapname"
    else
        echo "Snapshot already exists: $SNAPDIR/$snapname"
    fi
}

prune_snaps() {
    local prefix="$1"
    local keep="$2"

    mapfile -t snaps < <(find "$SNAPDIR" -maxdepth 1 -mindepth 1 -type d -name "${prefix}-*" | sort -r)

    if [ "${#snaps[@]}" -gt "$keep" ]; then
        for snap in "${snaps[@]:$keep}"; do
            rmdir "$snap"
            echo "Removed old snapshot: $snap"
        done
    fi
}

HOURLY_NAME="hourly-$(date +%F-%H)"
DAILY_NAME="daily-$(date +%F)"
WEEKLY_NAME="weekly-$(date +%G-W%V)"

# Hourly snapshots: keep 24
create_snap "$HOURLY_NAME"
prune_snaps "hourly" 24

# Daily snapshots: create at midnight, keep 14
if [ "$(date +%H)" = "00" ]; then
    create_snap "$DAILY_NAME"
    prune_snaps "daily" 14
fi

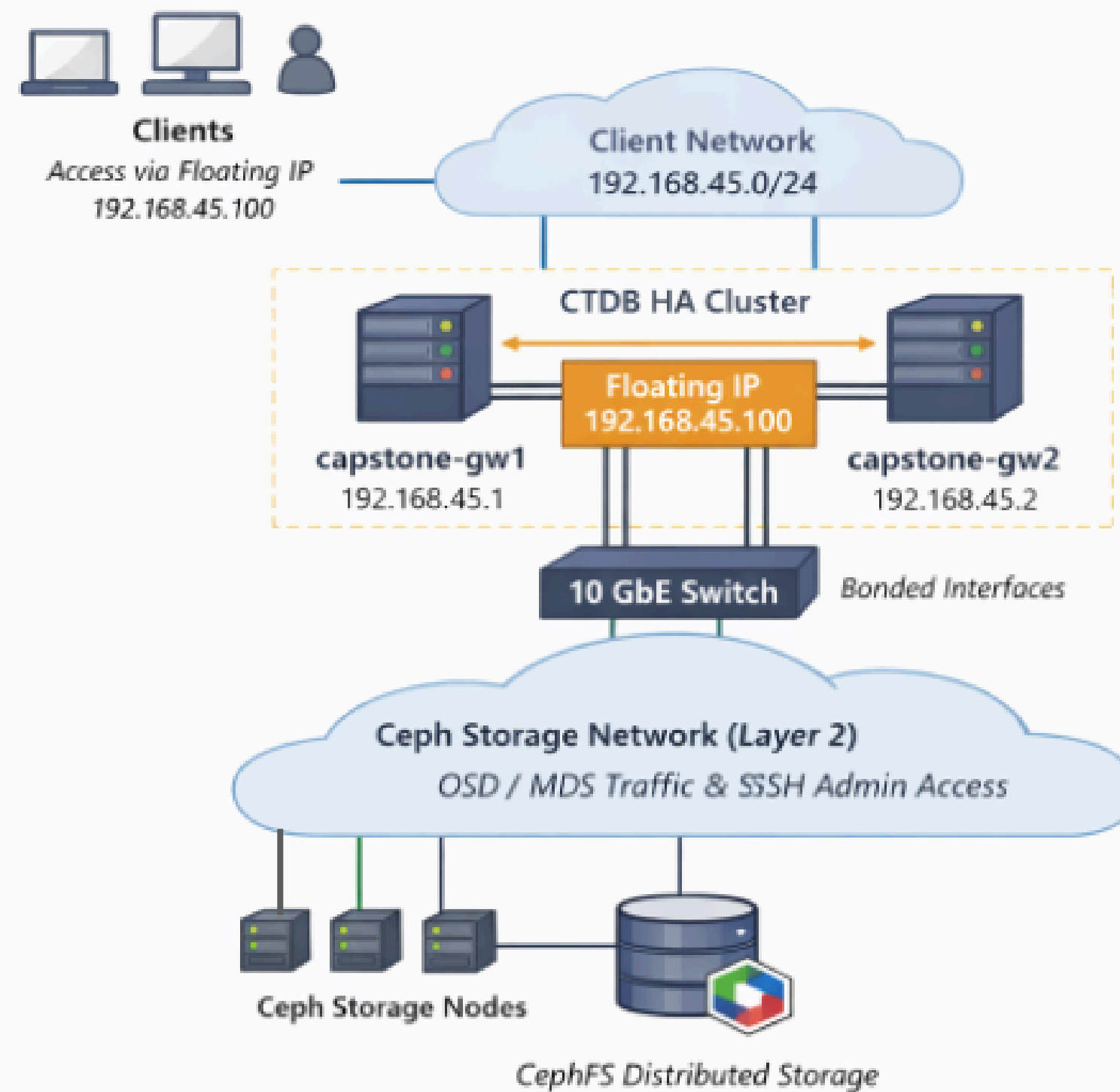
# Weekly snapshots: create Sunday at midnight, keep 8
if [ "$(date +%u)" = "7" ] && [ "$(date +%H)" = "00" ]; then
    create_snap "$WEEKLY_NAME"
    prune_snaps "weekly" 8
fi
```


SMB HA GATEWAY CLUSTER (CTDB)

- 2-node gateway cluster
 - Samba
 - CTDB
- Floating IP (192.168.45.100)
- Active/standby failover
- SMB share access

```
[root@capstone-gw1 ~]# ctdb status
Number of nodes:2
pnn:0 192.168.45.1      OK (THIS NODE)
pnn:1 192.168.45.2      OK
Generation:1925329388
Size:2
hash:0 lmaster:0
hash:1 lmaster:1
Recovery mode:NORMAL (0)
Leader:0
[root@capstone-gw1 ~]# ctdb ip
Public IPs on node 0
192.168.45.100 0
```

SMB Gateway Architecture



- Client Network
- Storage & Management Network
- Storage & Management Network

HA TESTING

```
//192.168.45.100/cephshare 71T 12M 71T 1% /mnt/cephshare  
[root@capstone-sn1 ~]# while true; do date >> /mnt/cephshare/testfile.txt; sle  
ep 1; done
```

root@capstone-sn1:~

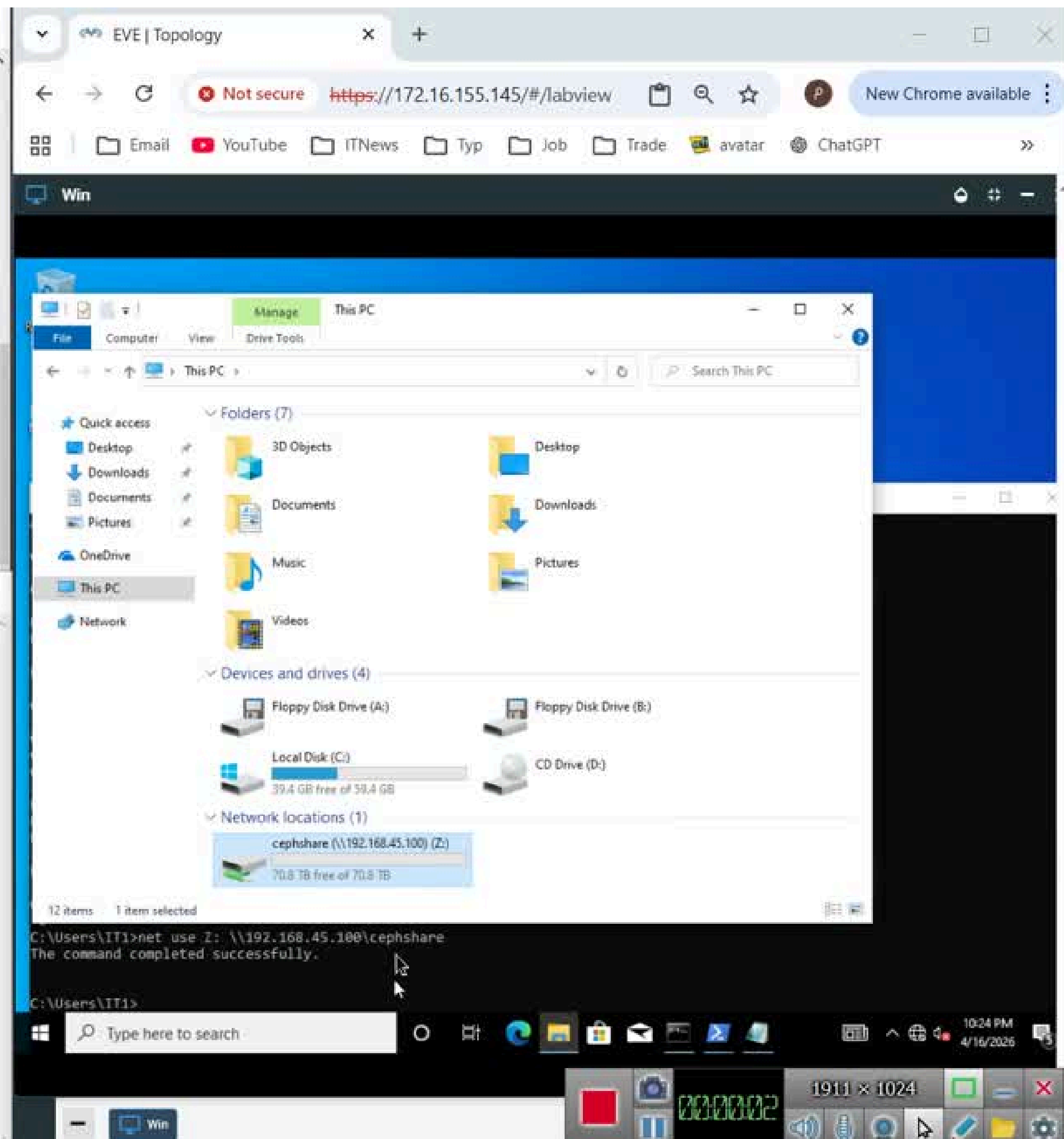
```
Thu 09 Apr 2026 03:58:19 AM ADT  
Thu 09 Apr 2026 03:58:20 AM ADT  
Thu 09 Apr 2026 03:58:21 AM ADT  
Thu 09 Apr 2026 03:58:22 AM ADT  
Thu 09 Apr 2026 03:58:23 AM ADT  
Thu 09 Apr 2026 03:58:24 AM ADT  
Thu 09 Apr 2026 03:58:25 AM ADT  
Thu 09 Apr 2026 03:58:26 AM ADT  
Thu 09 Apr 2026 03:58:27 AM ADT  
Thu 09 Apr 2026 03:58:28 AM ADT  
Thu 09 Apr 2026 03:58:29 AM ADT  
Thu 09 Apr 2026 03:58:30 AM ADT  
Thu 09 Apr 2026 03:58:31 AM ADT  
Thu 09 Apr 2026 03:58:32 AM ADT  
Thu 09 Apr 2026 03:58:38 AM ADT  
Thu 09 Apr 2026 03:58:39 AM ADT  
Thu 09 Apr 2026 03:58:40 AM ADT  
Thu 09 Apr 2026 03:58:41 AM ADT  
Thu 09 Apr 2026 03:58:42 AM ADT
```

Failover scenario:

1. Client writing data
2. Gateway node failure
3. CTDB failover
4. Client continues access

```
8: ens6f0: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc mq master bo
nd1 state UP group default qlen 1000
    link/ether a0:36:9f:3f:80:f8 brd ff:ff:ff:ff:ff:ff
    altname enp155s0f0
9: ens6f1: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc mq master bo
nd1 state UP group default qlen 1000
    link/ether a0:36:9f:3f:80:f8 brd ff:ff:ff:ff:ff:ff permaddr a0:36:9f:3f:80
:fa
    altname enp155s0f1
10: bond1: <BROADCAST,MULTICAST,MASTER,UP,LOWER_UP> mtu 1500 qdisc noqueue sta
te UP group default qlen 1000
    link/ether a0:36:9f:3f:80:f8 brd ff:ff:ff:ff:ff:ff
    inet 192.168.45.1/24 brd 192.168.45.255 scope global noprefixroute bond1
        valid_lft forever preferred_lft forever
    inet 192.168.45.100/24 brd 192.168.45.255 scope global secondary bond1
        valid_lft forever preferred_lft forever
    inet6 fe80::a036:9fff:fe3f:80f8/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[admin@capstone-gwl ~]$ sudo ctdb ip
Public IPs on node 0
192.168.45.100 0
```

```
[admin@capstone-gw2 ~]$ sudo ctdb status
Number of nodes:2
pnn:0 192.168.45.1      OK
pnn:1 192.168.45.2      OK (THIS NODE)
Generation:728093477
Size:2
hash:0 lmaster:0
hash:1 lmaster:1
Recovery mode:NORMAL (0)
Leader:0
[admin@capstone-gw2 ~]$
```

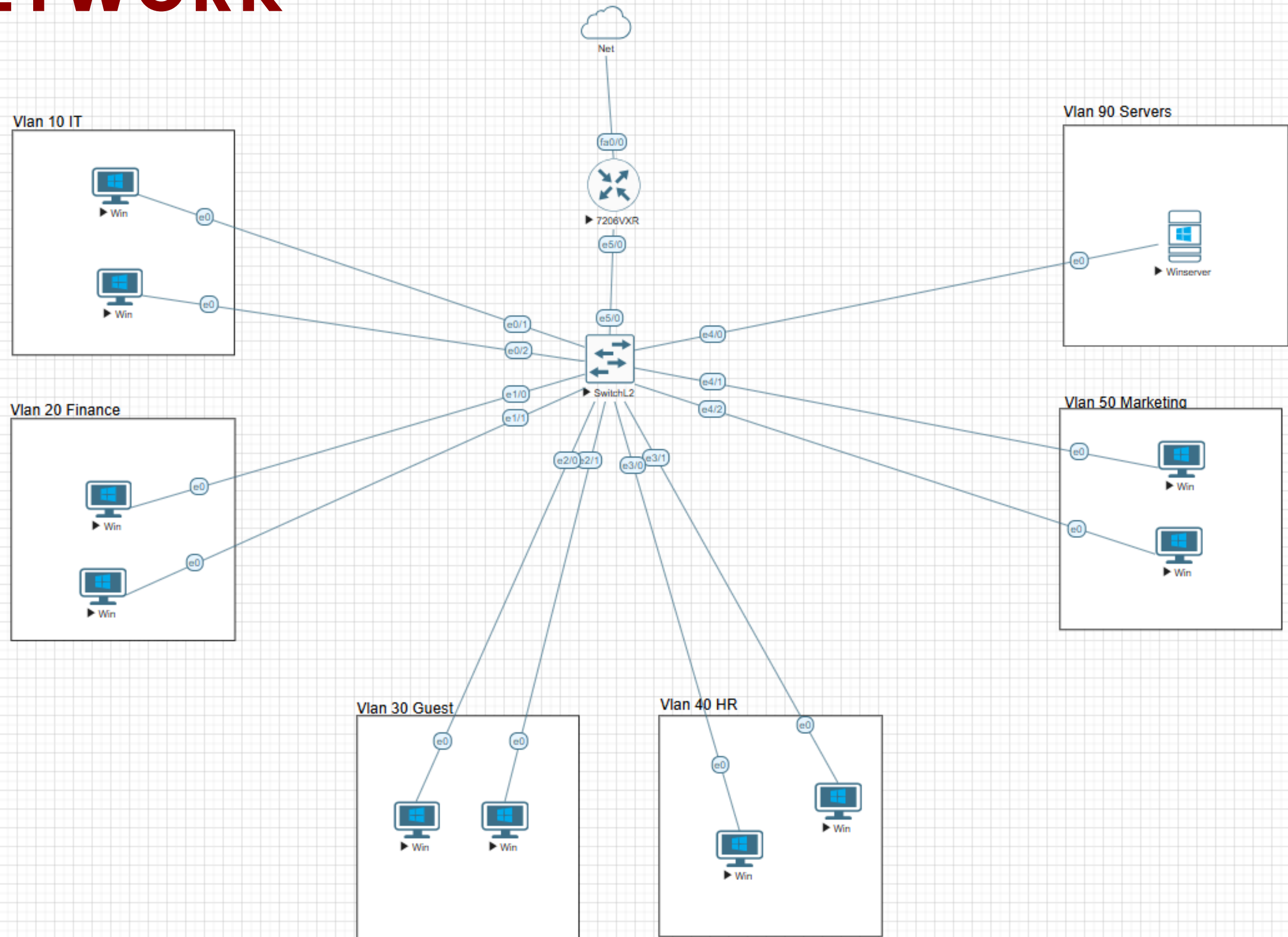


VIRTUAL OFFICE NETWORK (EVE-NG)

Simulated corporate network using EVE-NG

Components:

- Cisco Router
- Firewall
- Cisco Switches
- VLAN segmentation
- Multiple client PCs



SECURITY HARDENING

- No root login
- SSH key authentication
- Firewall rules (ansible)
- Network segmentation
- password policies

```
[admin@capstone-sn3 ~]$ su -  
Password:  
su: Authentication failure  
[admin@capstone-sn3 ~]$
```

```
PLAY RECAP *****  
**  
capstone-gw1      : ok=9    changed=5    unreachable=0    failed=0  
  skipped=0      rescued=0    ignored=0  
capstone-gw2      : ok=9    changed=3    unreachable=0    failed=0  
  skipped=0      rescued=0    ignored=0  
capstone-sn1      : ok=7    changed=1    unreachable=0    failed=0  
  skipped=1      rescued=0    ignored=0  
capstone-sn2      : ok=7    changed=2    unreachable=0    failed=0  
  skipped=1      rescued=0    ignored=0  
capstone-sn3      : ok=8    changed=0    unreachable=0    failed=0  
  skipped=0      rescued=0    ignored=0  
capstone-sn4      : ok=7    changed=2    unreachable=0    failed=0  
  skipped=1      rescued=0    ignored=0  
  
[admin@capstone-sn3 storage-cluster-admin]$ ls  
inventory.ini  lockdown.yml  
[admin@capstone-sn3 storage-cluster-admin]$ ansible-playbook -i inventory.ini  
lockdown.yml -K
```

Firewalld (Ansible)

- Zero-Trust Defaults: Switched all nodes to a drop policy to block all unauthorized traffic by default.
- Role-Based Zones: Isolated Storage Nodes (internal only) from Gateway Nodes (client-facing)
- Whitelisted Access: Used Rich Rules to grant storage access only to specific client subnets (10,20,40,50,90)
- Lockdown Mode: Disabled local firewall modifications via D-Bus to prevent unauthorized overrides
- Ansible Enforcement: Automated the configuration to ensure identical security across every node in the cluster

Challenges Encountered

- 1 Mon Time Sync Issue
- 2 CTDB high-availability cluster issue
- 3 Troubleshooting network connectivity inside the virtual office
- 4 EVE-NG



CONCLUSION

- Enterprise storage cluster deployed
 - Storage cluster tuned for better performance
 - Highly available SMB gateway
 - Virtual corporate network environment
 - Secure infrastructure design
- 

Q & A

**ANY
QUESTIONS?**





45Drives Capstone Project

THANK YOU